Mathematical Theory

Least Squares Method

Let X be the design matrix and let y be the observed data vector. The columns of X span the model subspace $\mathrm{Col}(X) = \mathrm{im}(X)$.

In general, the observed vector y does not lie exactly in $\mathrm{Col}(X)$. Therefore, we seek a vector of the form $X\beta$ in $\mathrm{Col}(X)$ that is as close as possible to y. This means finding the orthogonal projection of y onto $\mathrm{Col}(X)$.

Minimize $\|\boldsymbol{y} - \boldsymbol{X\beta}\|^2$

If $\hat{\beta}$ is a least squares solution, then the fitted vector and residual vector are

$$\begin{aligned} \hat{y} &= X\hat{\beta} \\ r &= y - \hat{y} = y - X\hat{\beta} \end{aligned}$$

Because $\hat{y}$ is the orthogonal projection of y onto $\mathrm{Col}(X)$, the residual vector is orthogonal to the column space of X; that is, $r \perp \mathrm{Col}(X)$. Since every column of X lies in $\mathrm{Col}(X)$, the residual is orthogonal to every column of X. Therefore,

$$\begin{aligned} X^T r &= 0 \\ X^T\left(y - X\hat{\beta}\right) &= 0 \\ X^T y - X^T X\hat{\beta} &= 0 \\ X^T X\hat{\beta} &= X^T y. \end{aligned}$$

These are the normal equations for the least squares problem.

If the columns of X are linearly independent, then X has full column rank and $\ker(X) = \{\vec{0}\}$. Since $\ker(X^T X) = \ker(X)$, it follows that $X^T X$ is invertible. Therefore, the unique least squares solution is

$$\hat{\beta} = (X^T X)^{-1} X^T y.$$

The fitted values are then

$$\hat{y} = X\hat{\beta}.$$

QR Method

Assume X has full column rank. Using the reduced QR-decomposition, we write

$$X = QR,$$

where the columns of Q are orthonormal, $Q^T Q = \mathrm{I}$, and R is an invertible upper triangular matrix. Substituting $X = QR$ into the normal equations gives

$$\begin{aligned} (QR)^T (QR)\hat{\beta} &= (QR)^T y \\ R^T Q^T (QR)\hat{\beta} &= R^T Q^T y \\ R^T R\hat{\beta} &= R^T Q^T y. \end{aligned}$$

Since R is invertible, we can reduce the system to

$$R\hat{\beta} = Q^T y.$$

Thus, in the QR method, the least squares coefficients are found by solving an upper triangular system. This can be done efficiently using back-substitution.